Deep Learning Sentiment Analyzer on Albanian-language Social Media

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Abstract

Albanian, spoken by 10 million speakers, faces challenges as a low-resource language in natural language processing (NLP). This paper addresses this gap by contributing to sentiment analysis architectures in Albanian, using a dataset of Facebook comments related to the COVID-19 pandemic. Our study evaluates the performance of various deep learning models, including the Bidirectional Long Short-Term Memory (BiLSTM) with attention mechanism, Multilingual BERT (mBERT), and Cross-lingual Language Model RoBERTa (XLM-R). Results show that XLM-R performs the best overall, followed by mBERT with stop word removal. Interestingly, fastText embeddings outperform other embeddings in the BiLSTM models. We discuss the implications of our findings and highlight the importance of further research in developing NLP tools for Albanian.

1 Introduction

Albanian is spoken by 10 million speakers and it is the national language of Albanian and Kosovo, two Balkan states. As a low-resource language, there is a lack of NLP resources for Albanian. NLP tools crucially enable speakers of low-resource languages to use digital tools like the Internet or AI in their native language. They also serve as the foundation for developing various language technologies, such as machine translation, speech recognition, and text-to-speech synthesis. This project aims to contribute to deep learning sentiment analysis architectures in Albanian.

Sentiment analysis, also known as opinion mining, is a NLP technique that involves determining the emotion expressed in a text. Its objective is to classify the sentiment conveyed in the text as positive, negative, or neutral. It is a useful tool to gauge public opinion. The database we employed is a collection of Facebook comments related to the COVID-19 pandemic. The sentiment analysis of social media data can give decision makers in a variety of fields (politics, marketing, public relations) a direct channel to social media users' feelings on a subject.

Our main contributions are:

- We show XLM-R performs the best of all models to predict sentiment analysis on this task.
- We show mBERT performs better on our task *with* stop word removal as part of its text preprocessing.
- We revise the accuracy of fastText embeddings found by the original researchers (Kastrati et al., 2021), and assert that they produce the best results of all of the BiLSTM with attention mechanism models.
- We release all code on Github. [1].

2 Related Work

Low resource languages are languages that lack essential digital linguistic resources, such as large corpora, NLP tools, or well-trained language models specific to their language. According to Kastrati and Biba's 2022 state-of-the-art survey on Albanian NLP tools, "research works investigating NLP for the Albanian language are still limited." Compounding this, Albanian is a language isolate within the Indo-European language family, which poses a challenge for cross-lingual transfer learning from higher-resource Indo-European languages like English, Spanish, French or German.

Fortunately, Albanian is covered in two pretrained multilingual transformer models which we use for opinion mining in this work. The two transformer models are the Multilingual Bidirectional Encoder Representations from Transformers (mBERT) released by Devlin et al. (2019) and

[1](https://github.com/puopolo/DL-Albanian-sentiment, accessed on 22 March 2024)

Cross-lingual Language Model RoBERTa (XLM-R), published by Conneau et al. (2020). Whether mBERT or XLM-R performs better is disputed; results change depending on the language and task. Ács et al. (2021) compared the performance of mBERT and XLM-R across nine languages. They found that XLM-R outperformed mBERT in most morphological and part of speech (POS) tagging tasks, while mBERT has an advantage at named entity recognition (NER) tasks and at all French tasks. Meanwhile, Prifti Skenduli & Grabocka (2018) found that XLM-R out-performed BERT in a sentiment analysis task on microblogs in Albanian. Both mBERT and XLM-R are remarkable for including as many low-resource languages, though they still require work (Wu and Dredze, 2020).

3 Data

The AlbAna dataset (Kastrati et al., 2021) consists of 10,742 manually classified Facebook comments related to the COVID-19 pandemic. The Facebook comments were pulled from the National Institute of Public Health of Kosova Facebook page between March and August 2020. The AlbAna dataset is accessible on Github[2]. Each comment is labeled as neutral (0), positive (1), or negative (2). For each Facebook comment in the dataset, it lists: the number of likes it received, the comment's and post's timestamps, the post's URL, some Covid-related statistics, the annotation assigned by each of the three human annotators (0,1,2), and the final annotation (decided by majority). For this work, we removed all of the columns except for the comment and the final annotation. We use all 10,742 comments in this analysis.

3.1 Pre-processing

Pre-processing is a crucial step in text processing that involves cleaning and transforming raw text data into a format better for data analysis. Common pre-processing steps for NLP include tokenization, lowercasing, stop word removal, stemming, lemmatization, spell checking, and feature engineering. Kastrati et al. (2021) tokenized and lowercased their dataset, and applied term frequency inverse document frequency, *tf • idf*.

3.1.1 Word embeddings

For word embeddings, Kastrati et al. (2021) employed fastText, GloVe, and GloVe for twitter, and with mBERT they used its pre-trained embeddings. To build on this work, we only introduce one extra pre-processing step: stop word removal. We employed fastText and Glove embeddings with the BiLSTM architecture, BERT embeddings with mBERT, and XLM-R embeddings with XLM RoBERTa architecture. GloVe embeddings use statistics of a word's frequency and collocations across the whole text corpus to create embeddings. In contrast, fastText is a lightweight embedding model that treats each word as composed of n-grams, which gives it the advantage of easily processing rare words and unknown vocabulary. The BERT and XLM-R models use subword tokenization using byte-pair encoding (BPE) to mitigate unknown words, long words, and large vocabularies.

4 Deep Learning architectures

Four main deep learning classification models were employed by Kastrati et al. (2021): a bidirectional long short-term memory (BiLSTM) model, a 1D convolutional neural network (1D-CNN), a hybrid model, and BERT. They found the BiLSTM model with the attention mechanism performed the best, so in this work we only employ this model in comparison to Multilingual BERT (mBERT) and Cross-lingual Language Model RoBERTa (XLM-R), two multilingual transformers.

4.0.1 BiLSTM

Bidirectional Long Short-Term Memory, or BiLSTM, is a type of recurrent neural network (RNN). It is bidirectional because it passes information forward and backwards, so that every component of an input sequence is contextualized by both previous inputs and the present input. It is also good for long range dependencies.

4.0.2 Attention mechanism

An attention mechanism is a tool that highlights the importance of some features of the data inside a model. It assigns higher weights to important parameters, so that the model can focus on some specific inputs and better capture patterns in the data. Kastrati et al. (2021) calculated this task's ideal window size as 8, so that remains constant in this work.

4.0.3 Multilingual transformers

This work employs two pre-trained mutilingual transformers in the BERT family, Mutilingual

[2](https://github.com/lule-ahmedi/AlbAna, accessed on 18 March 2024)

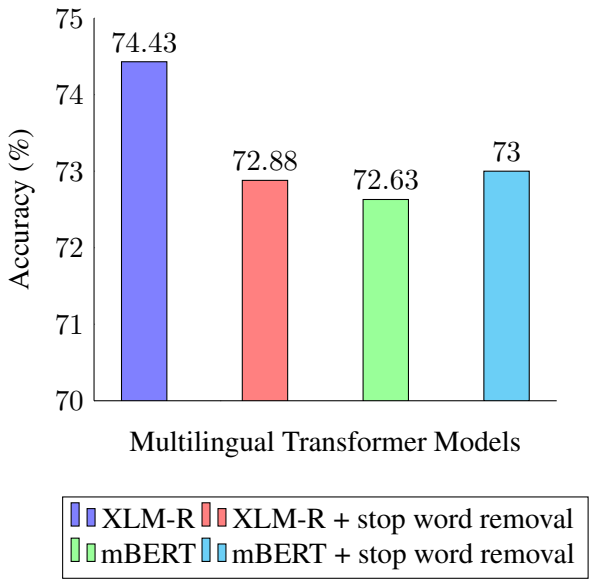

Figure 1: Comparison of Accuracy scores overall for the pre-trained multilingual transformer models.

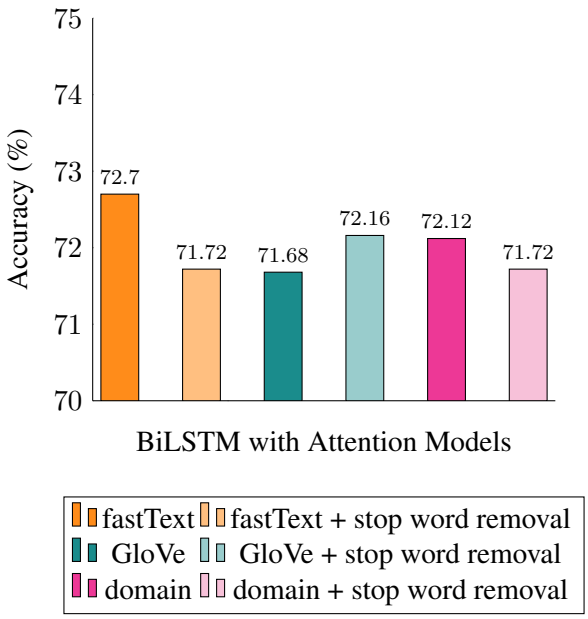

Figure 2: Comparison of accuracy scores for the BiLSTM models

BERT (mBERT) and Cross-lingual Language Model RoBERTa (XML-R). Both models are trained on data from 100+ languages, including Albanian. XLM-R was published a year after BERT and was pre-trained on more data.

4.1 Model performance

The pre-trained multilingual transformer models mBERT and XML-R performed better overall than the BiLSTM models. As Figure 1 shows, XLM-R (without stop word removal) has the highest accuracy of any model, followed by mBERT with stop word removal. As the appendix Table 3shows, XLM-R (without stop word removal) has the highest accuracy and the highest F1-scores for neutral and positive classes, but XLM-R with stop word removal and mBERT (as is) better predict at the sentiment of negative comments.

4.2 Word embeddings

Among the BiLSTM models, fastText outperforms all of the other embeddings. The next highest accuracy among the BiLSTM models using the GloVe embeddings with stop word removal. The average accuracy of each BiLSTM model is displayed in Figure 2.

4.3 Stop word removal

Adding stop word removal as a pre-processing step had a small but varying effect. In the XLM-R model and the BiLSTM models with fastText and domain embeddings, stop word removal had a negative effect on their average accuracy. However, the accuracy of the mBERT model and the BiLSTM model with GloVe embeddings was improved by stop word removal.

5 Discussion

Results changed by much less than we expected. Given the class imbalance of neutral comments (56.4%), this is the baseline accuracy because even a dumb classifier that labeled every comment as Neutral would get 56.4% correct. The confusion matrixes for all models are visible in the Appendix, and it is encouraging that none of them do this. One possible reason that fastText performed the best is because it was trained on Facebook data. Why these results differ from (Kastrati et al., 2021)'s though, is unclear. It is feasible that fastText has updated its technology in the last couple years since their research. It is also unclear to us why stop word removal had a varying effect. Perhaps this pre-processing step had a negative effect on the XLM-R model's accuracy because it was pre-trained with so much data that it depends on stop words more than mBERT, for example, which slightly benefited from the stop word removal step. Only with more publicly-available NLP tools for Albanian will we have a clearer picture of whether linguistic-based pre-processing steps actually improve or impair deep learning models.

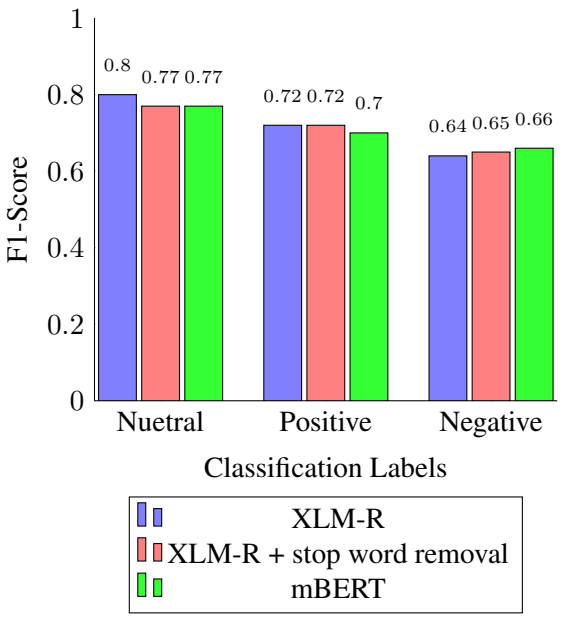

Figure 3: Comparison of F1-Scores for the top performing models by class

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A Appendix

Table 1: BiLSTM with Attention Mechanism Performance Metrics

Embeddings	**Metrics**			
	Precision	**Recall**	**F1-score**	**Support**
fastText	0.73	0.87	0.79	1268
	0.78	0.58	0.67	358
	0.69	0.53	0.59	630
	Accuracy: 0.7270			
fastText + Stop word removal	0.76	0.77	0.77	1268
	0.78	0.61	0.68	358
	0.60	0.66	0.63	630
	Accuracy: 0.7172			
GloVe	0.73	0.84	0.78	1268
	0.77	0.60	0.67	358
	0.65	0.54	0.59	630
	Accuracy: 0.7168			
GloVe + Stop word removal	0.74	0.82	0.78	1268
	0.75	0.62	0.68	358
	0.65	0.58	0.62	630
	Accuracy: 0.7216			
Domain	0.73	0.85	0.78	1268
	0.77	0.58	0.66	358
	0.67	0.53	0.59	630
	Accuracy: 0.7212			
Domain + Stop word removal	0.74	0.81	0.78	1268
	0.77	0.58	0.66	358
	0.64	0.61	0.62	630
	Accuracy: 0.7199			

Table 2: Multilingual Transformer Performance Metrics

Model	**Metrics**			
	Precision	**Recall**	**F1-score**	**Support**
XLM-R	0.79	0.81	**0.80**	1837
	0.70	0.74	**0.72**	507
	0.67	0.62	0.64	879
	Accuracy: 0.7443			
XLM-R with Stop word removal	0.80	0.75	0.77	1837
	0.78	0.67	**0.72**	507
	0.59	0.72	**0.65**	879
	Accuracy: 0.7288			
mBERT	0.83	0.73	0.77	1837
	0.69	0.72	0.70	507
	0.59	0.73	**0.66**	879
	Accuracy: 0.7263			
mBERT with Stop word removal	0.79	0.77	0.78	1837
	0.86	0.60	0.70	507
	0.59	0.71	0.64	879
	Accuracy: 0.7300			

A.1 Confusion Matrices

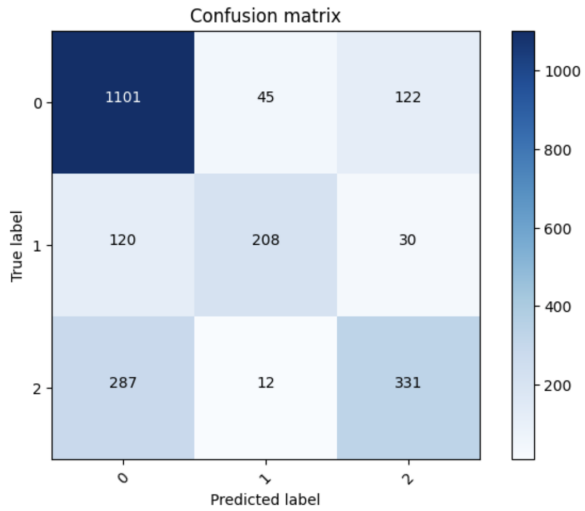

Figure 4: BiLSTM with attention mechanism and fastText embeddings

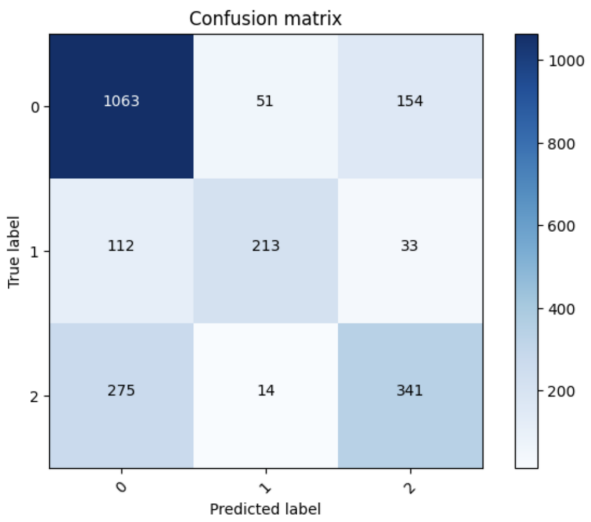

Figure 5: BiLSTM with attention mechanism and GloVe embeddings

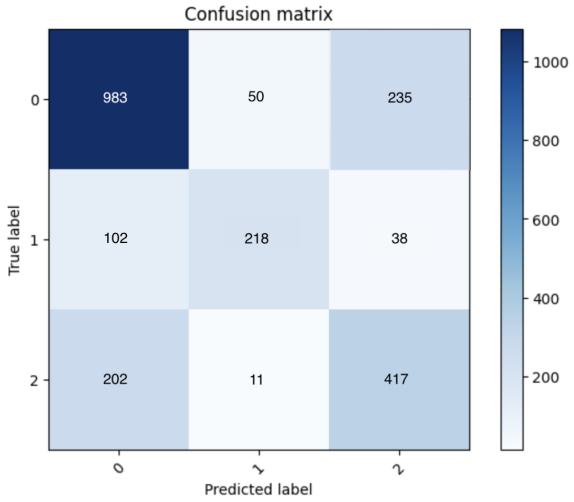

Figure 6: BiLSTM with attention mechanism, fastText embeddings, and stop word removal

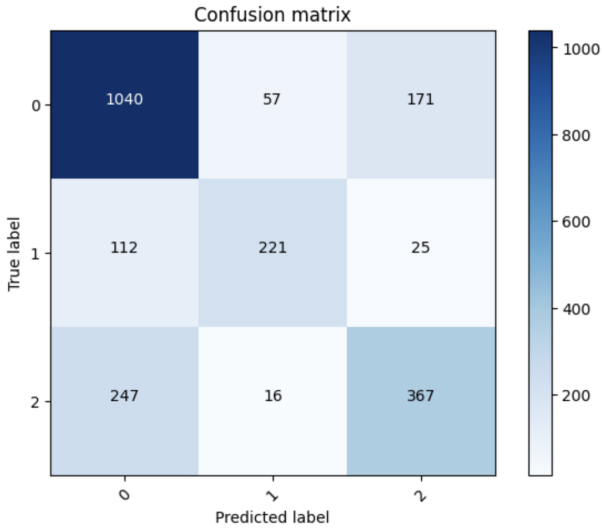

Figure 7: BiLSTM with attention mechanism, GloVe embeddings, and stop word removal

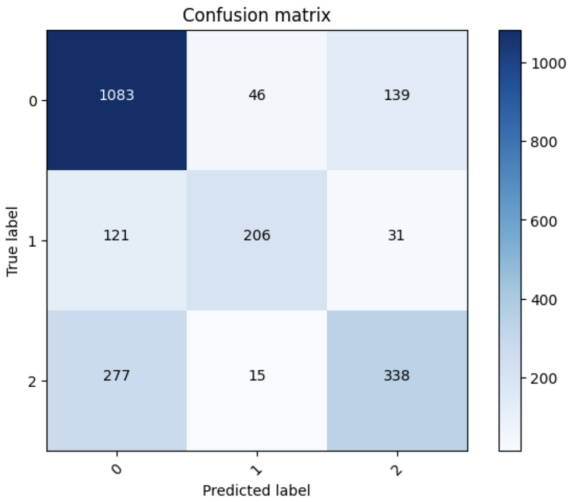

Figure 8: BiLSTM with attention mechanism and domain embeddings

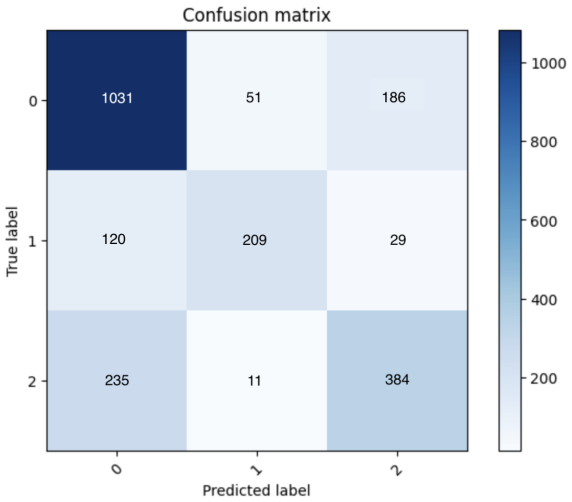

Figure 9: BiLSTM with attention mechanism, domain embeddings, and stop word removal

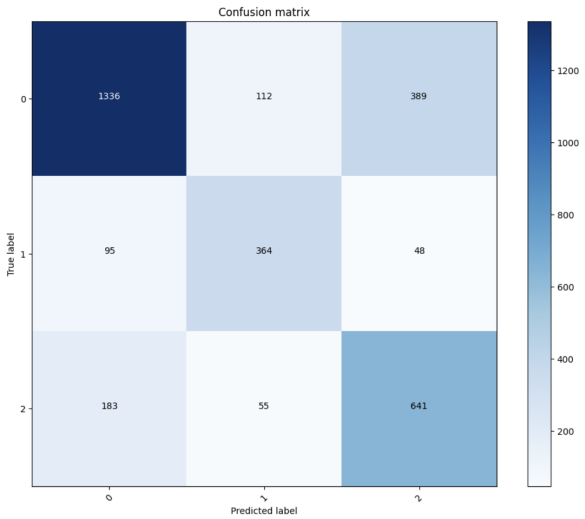

Figure 10: mBERT

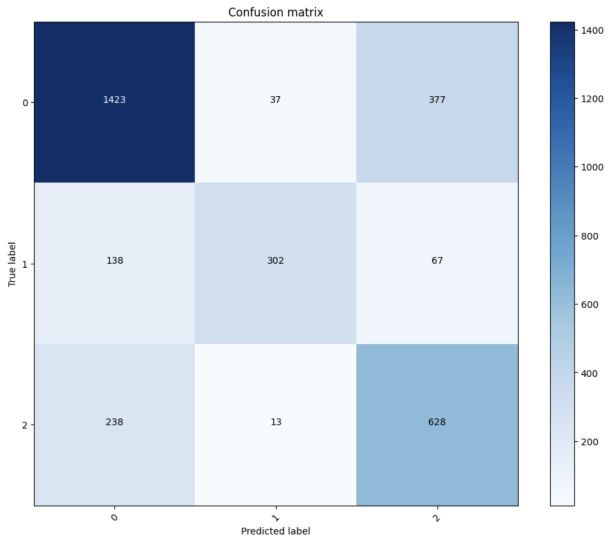

Figure 11: mBERT with stop word removal

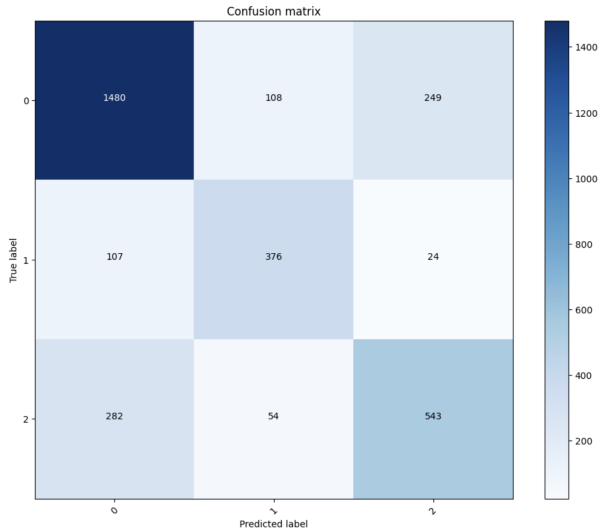

Figure 12: XLM-R

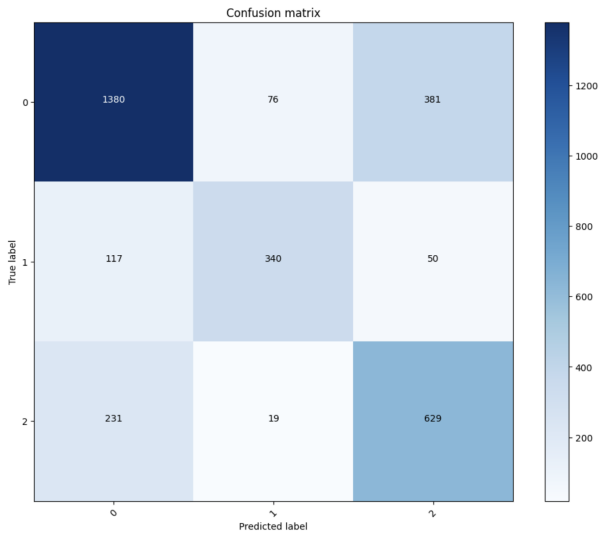

Figure 13: XLM-R with stop word removal